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VCG - L2 - SOE/CXP

View only

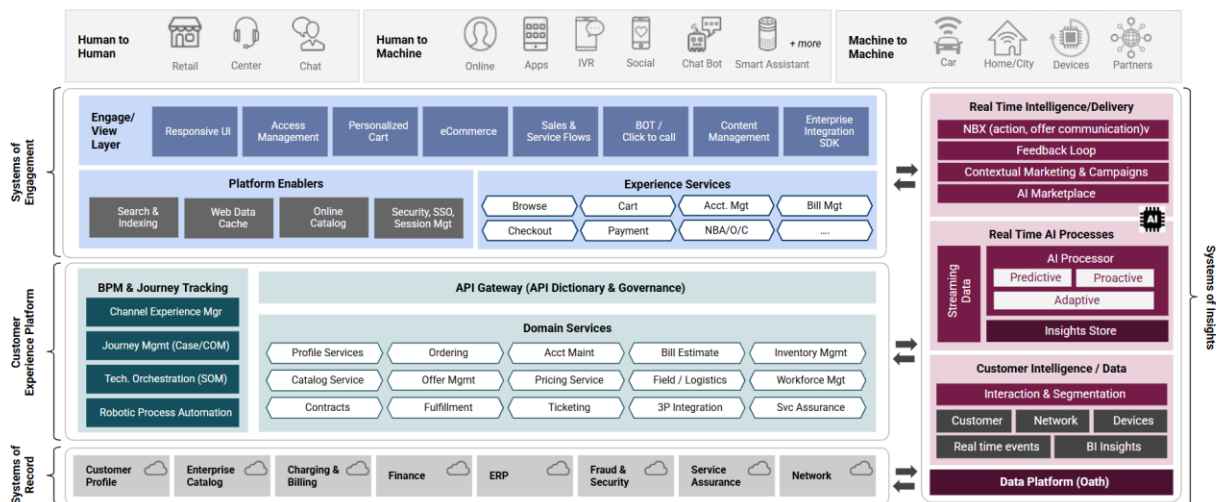
Reference Architecture Guiding Principles

- ★ **Architecture Foundational Areas** >>> **Systems of Engagement, Insights & Records**
- ★ **Common Digital Experience Platform** >>> **Uniform engagement & BPM processes across SOE & SOR**
- ★ **Customer-centric over Channel-centric** >>> **Consistent experience across all touchpoints**
- ★ **Common SOI tier across all channels** >>> **Oath/VCG grid baseline, enabling a 1:1 customer experience**
- ★ **Enable consistent, reusable services** >>> **API first approach, with reusable design patterns**
- ★ **Efficient use of resources** >>> **Standard technology stack to create a interchangeable resource pool**

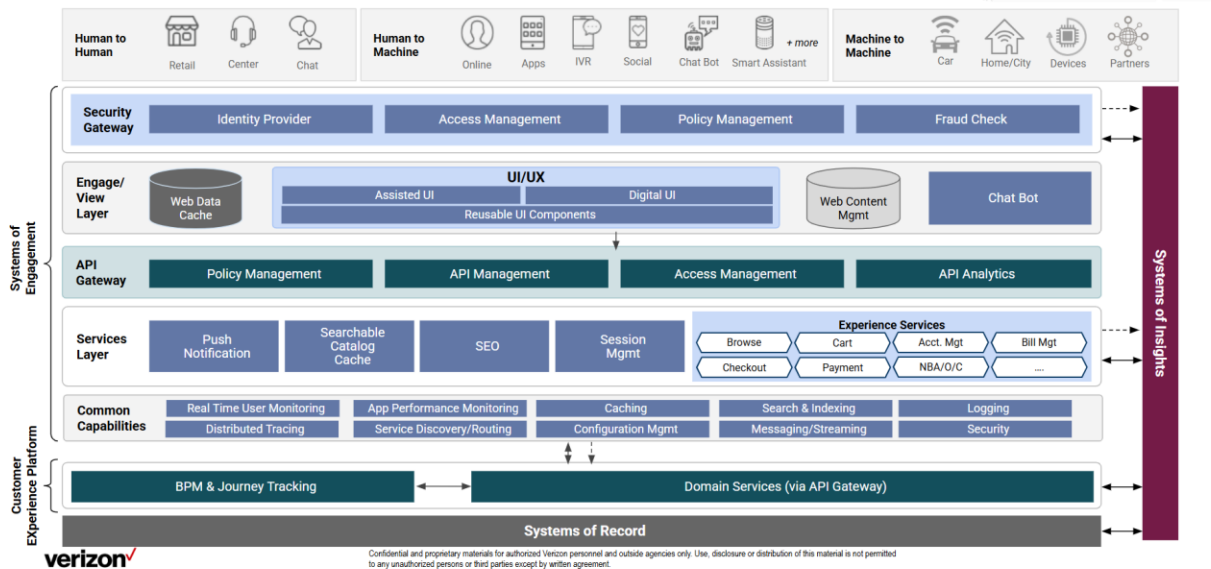
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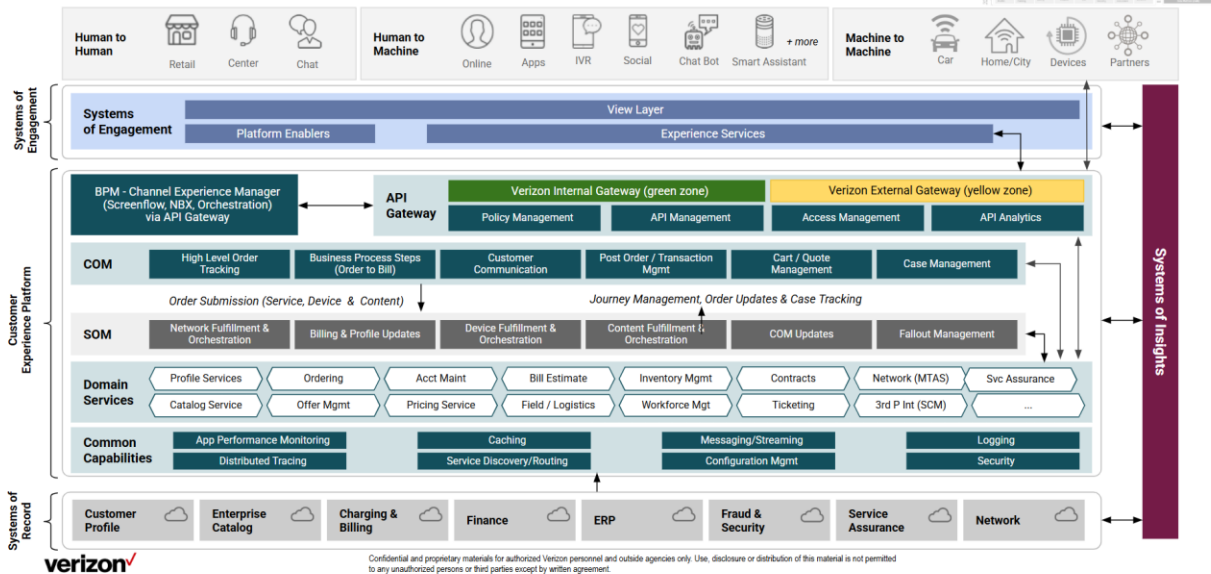
North Star: End State Reference Architecture



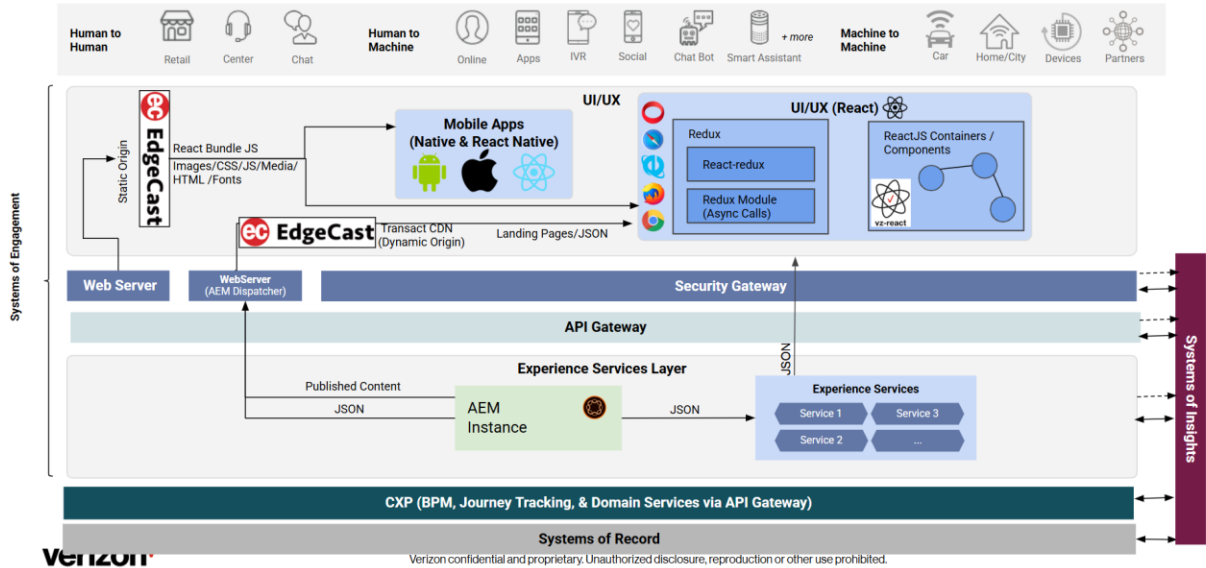
Systems of Engagement (Level 1 Architecture)



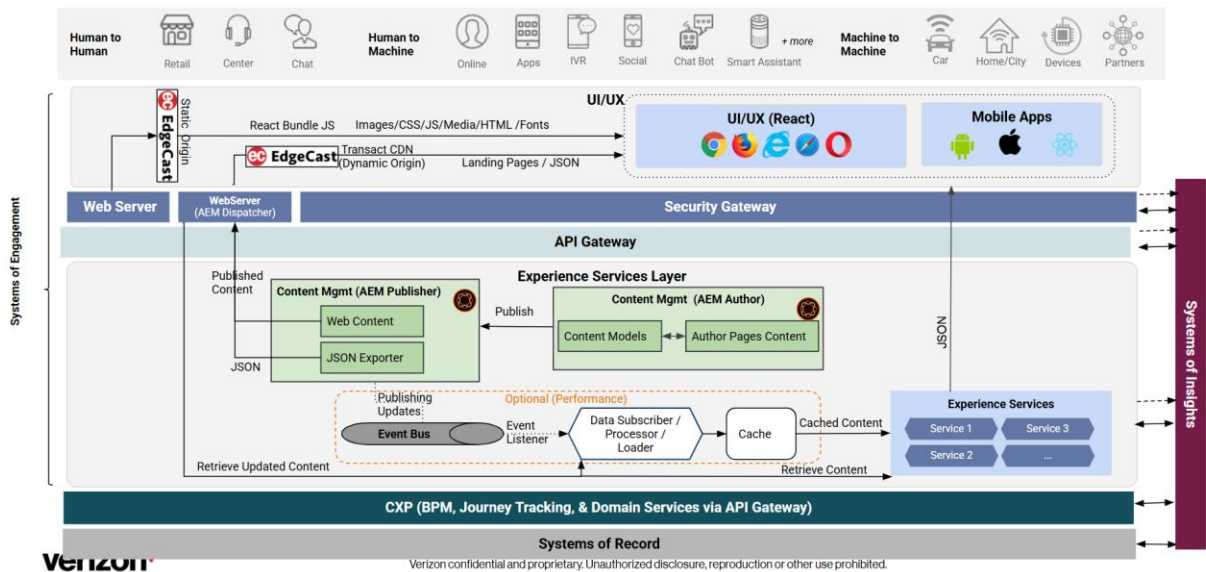
Customer Experience Platform (Level 1 Architecture)

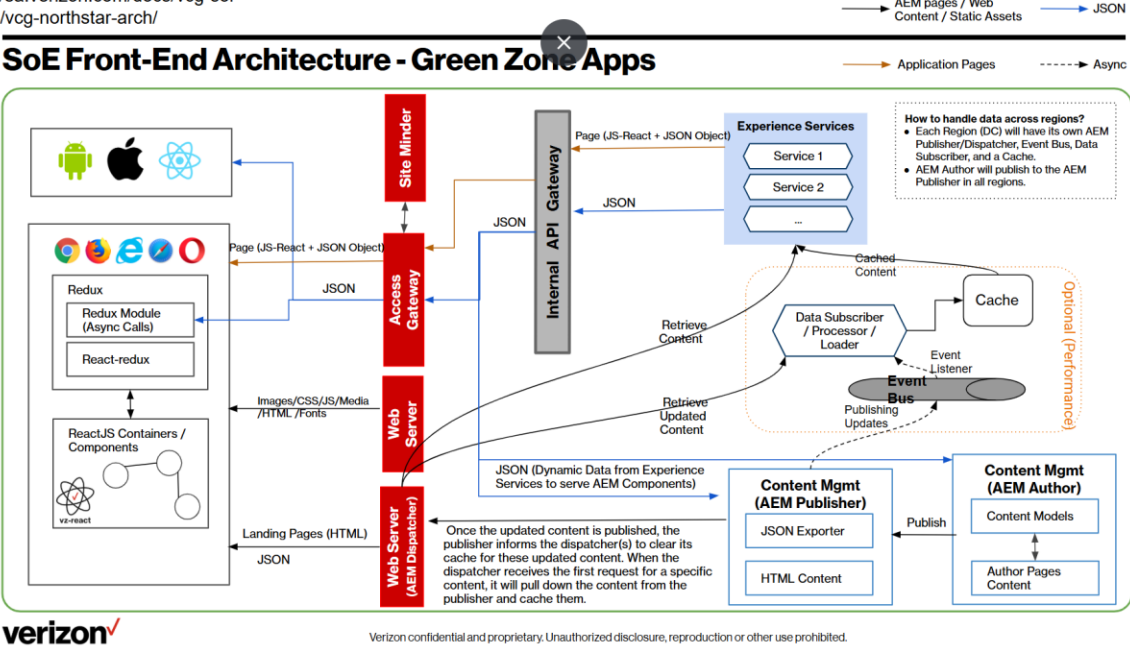


SoE Front-End Architecture



SoE Static Web Content Architecture





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SoE Front-End Architecture - Target

SPA denotes Single Page Application
MPA denotes Multiple Page Application

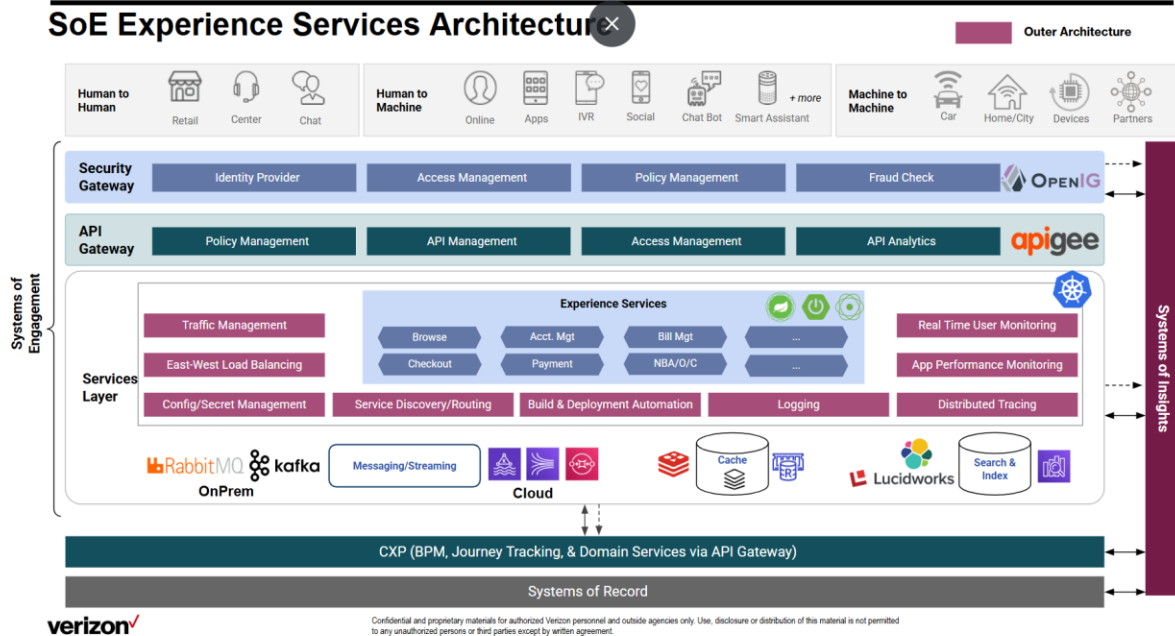
Capabilities	SoE Web Target	SoE Mobile App Target
UI Framework	ReactJS	Native / React Native*
Rendering	Client-Side	Client-Side
Architecture	Multiple SPAs	Multiple SPAs
WCM - Web Content Mgmt	AEM 6.4	AEM 6.4
SEO	Multiple Landing Pages for SEO purposes	N/A

*React Native is NOT a silver bullet and will not address all your needs. Please evaluate based on your app requirements. RN is quickly evolving

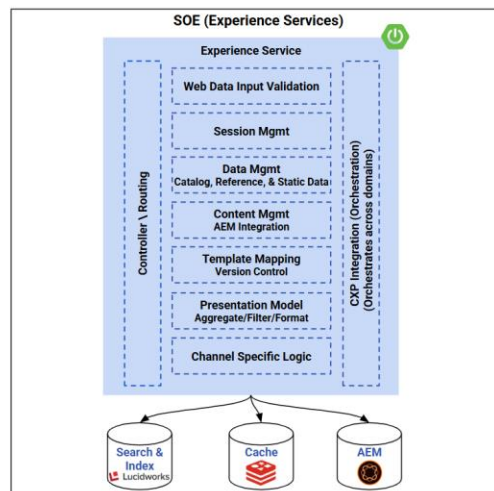


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SoE Experience Services Architecture



Experience Service - Anatomy



SoE Experience Services & Outer Arch - Target Technologies

Capabilities	SOE Target Technologies
Experience Services	Spring Boot 2.x / Spring 5.x / Spring Reactor
Logging	EFK (Elastic, FluentD, Kibana/Grafana)
Caching	Redis
App Performance Monitoring	NewRelic
Distributed Tracing	NewRelic
Real Time User Monitoring	Glassbox
Platform / Deployment	AWS/Kubernetes
Messaging / Streaming	SQS/Managed Kafka/Kinesis (AWS) & RabbitMQ/Kafka (On Prem)
Search & Index	Lucidworks
CI/CD	Jenkins
Service Discovery	K8s DNS, K8s Service
East-West Load Balancing	K8s Service
Configuration Management	K8s Config Map
Secret Management	K8s Sealed Secrets
Service Routing / North-South Load Balancing	K8s Ingress Controller
Outbound Traffic Control	K8s Egress



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Access Management Use Cases & User Types

AuthN/AuthZ use cases:

- **VZ Applications that are internet facing (Internet - Yellow Zone - Green Zone):**
 - Digital Web Applications
 - Digital Mobile Applications
 - Assisted - Indirect Reps (e.g. Costco)
- **VZ applications within the Green Zone**
 - Assisted - VZ Reps (including ACSS)
- **External APIs - 3rd Party Integrations (External API Gateway) (Internet - Yellow Zone - Green Zone):** e.g. Target & BestBuy
- **Internal APIs (Internal API Gateway)**
 - Experience APIs consumed by Internal VZ Application
 - Domain APIs consumed by Experience Services
- **Federated Identity - VZ Customers authenticating thru 3rd party websites/apps (bestbuy.com)**
- **VZ-managed application/technology outside of the VZ network in a SaaS model (salesforce).**
- **IVR**
- **Citrix - External Telesales**
- **VPN Access - M2M/IoT**

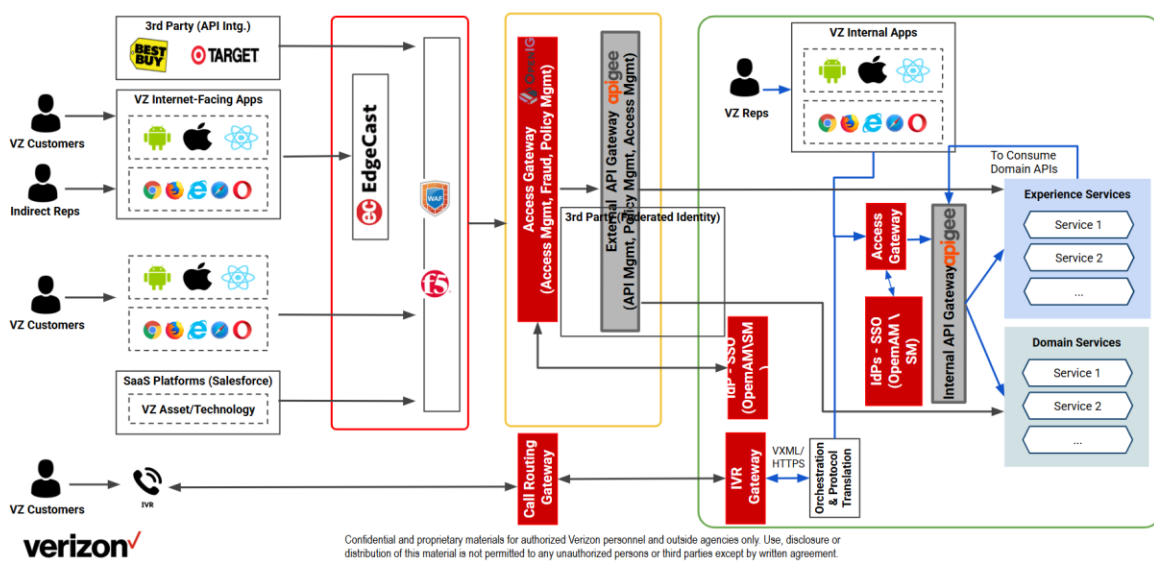
User Types:

- Customers
- VZ Reps (Employees)
- Indirect Reps (3rd party reps utilizing Verizon front-end assets and authenticating against SiteMinder)
- 3rd party partners (3rd party systems consuming verizon APIs)
- External Telesales

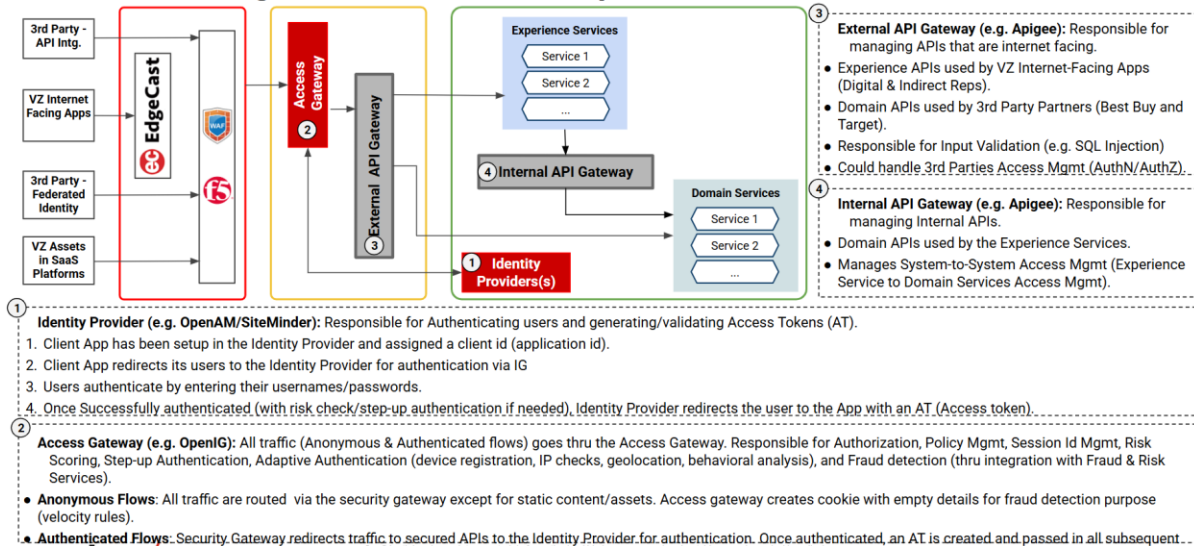


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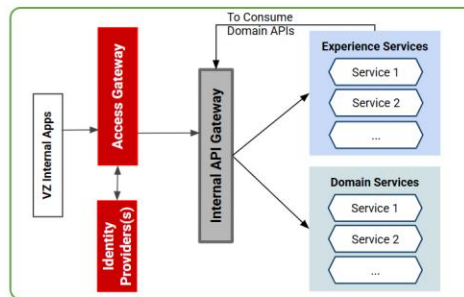
Access Management Use Cases & User Types



Access Management - Functional Capabilities - External Traffic



Access Management - Functional Capabilities - Internal Traffic

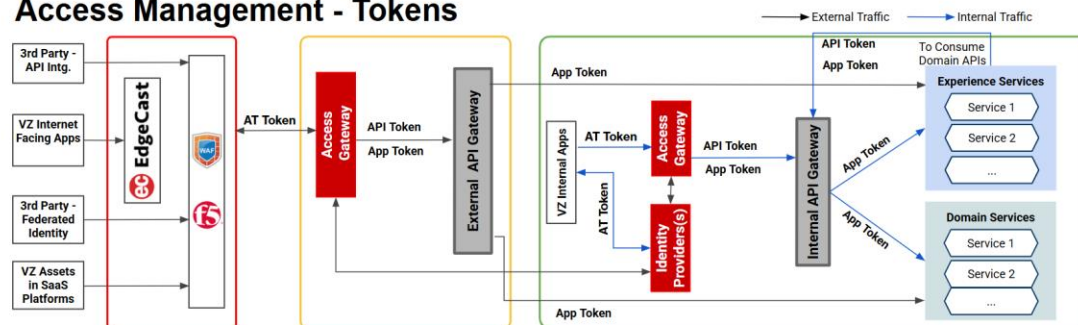


- Just like Internet-facing Traffic, similar capabilities are needed for Internal traffic (Green-Zone Apps).
- Internal API Gateway exposes Experience APIs and Domain APIs.
 - Experience APIs are consumed by the VZ Internal Apps.
 - Domain APIs are consumed by the Experience Services.



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Access Management - Tokens

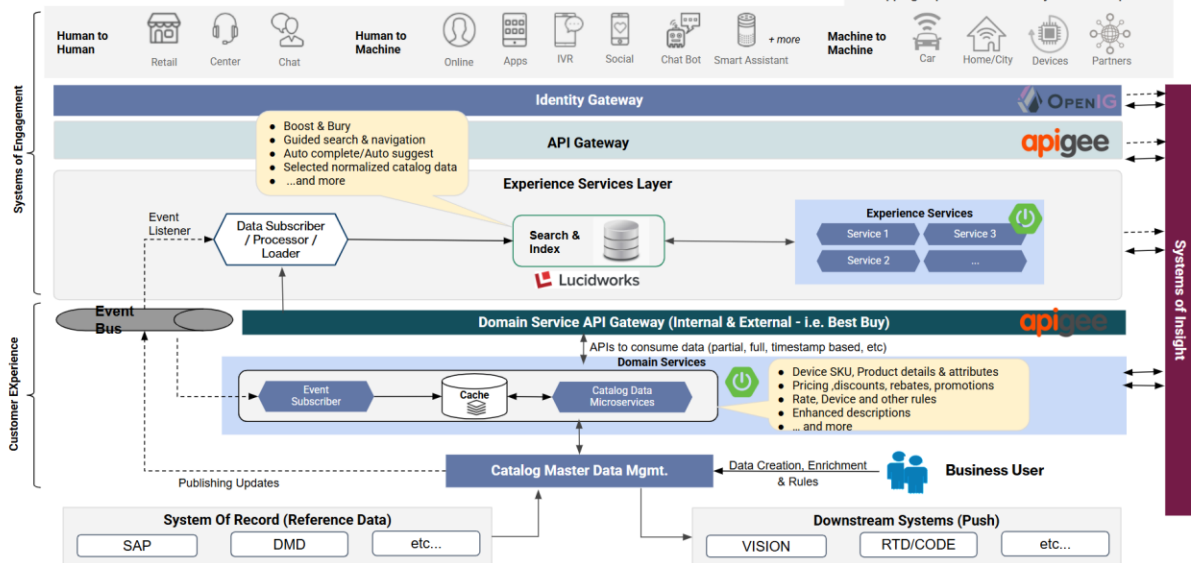


- **Access Token (AT):** Once a user is authenticated, IdP (OpenAM or SM) will generate an Access Token linked to the user account. Two Types:
 - **Opaque Token:** Opaque Tokens contain some identifier (GUID) to information in a server's persistent storage. To validate an opaque token, the recipient of the token needs to call the server that issued the token. For Authorization, the identifier (GUID) has to be passed to the IdP (OpenAM or SM) for validation and to retrieve any required user related data elements.
 - **JWT Token:** The JWT token contains encrypted user-related data elements and is signed by the IdP. For Authorization, the token signature can be verified (utilizing the IdP public key) and the data elements can be decrypted (what type of key is used for data elements encryption/decryption?).
- **Refresh Token (RT)** is used to refresh the Access Token (AT).
- **API Token (Apigee Token):** A token generated by the API Gateway (utilizing Apigee's Client ID and Secrets) for a specific API consumer (system/application not a user) and is used to authorize API calls within the API Gateway. API Tokens are used for Access Gateway → API Gateway integration and for Experience Services → Domain Services integration. In certain cases where the API consumer cannot perform token generation, basic auth headers (utilizing Apigee's Client ID and Secrets) over **two-way SSL** mechanism is utilized.
 - Access Gateway → API Gateway: Both Access Gateway and API Gateway are federated platforms so each application onboarded to the Access Gateway should get its own API token in order achieve fine grained control on the list of APIs that each app can consume. (This way not all apps configured within the AG have access to the same set of APIs in the API Gateway)
- **App Token (IG Token):** Access Gateway packages the needed data elements (e.g. account, mdn, user id, rep id, or user role) into an App Token (signed by Access Gateway) so it can be passed all the way down to all the needed systems (experience and domain services) to securely carry forward the business context. App tokens are validated at the API Gateway.

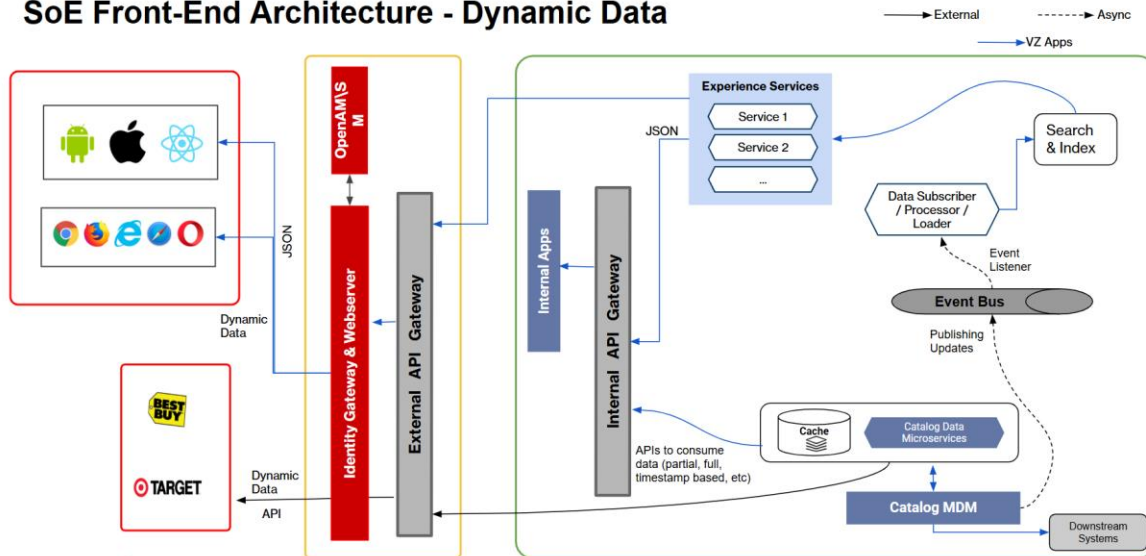


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Search & Cache Architecture for Dynamic “Catalog” Data



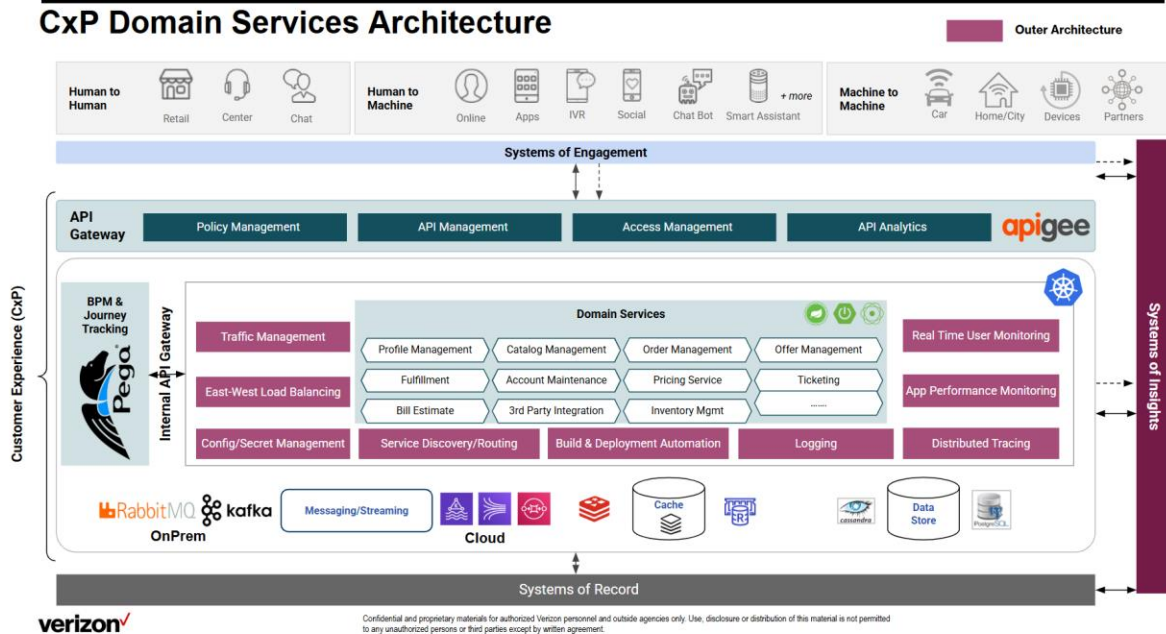
SoE Front-End Architecture - Dynamic Data



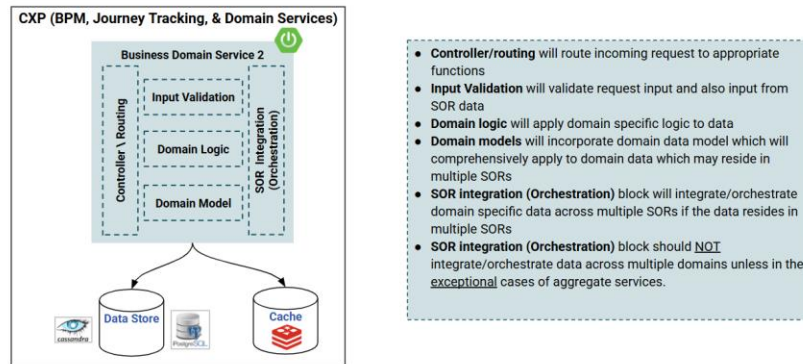
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CxP Domain Services Architecture



Domain Services - Anatomy



Domains – Identified

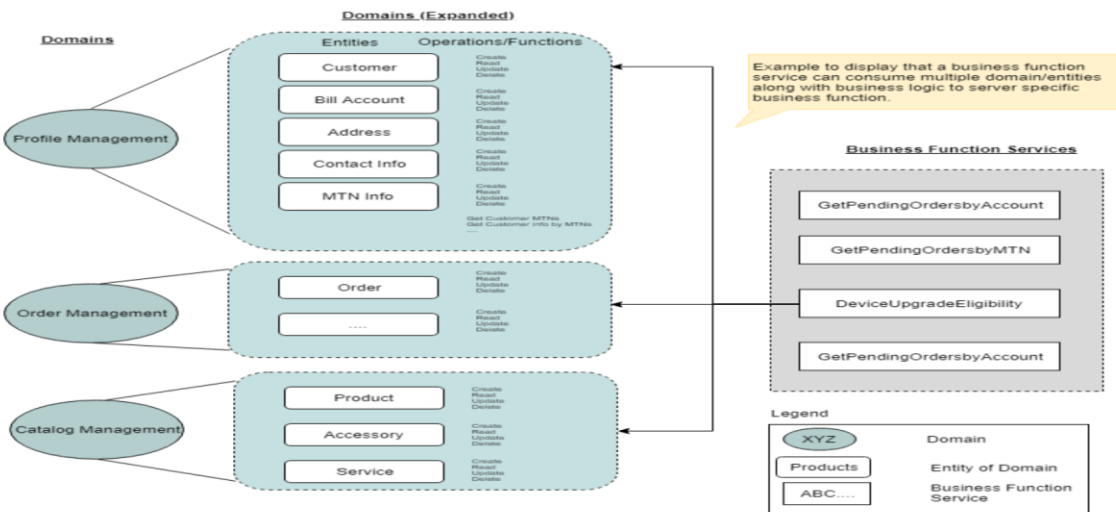
- Profile Management
- Catalog Management
- Order Management
- Offer Management
- Fulfillment
- Account Maintenance
- Pricing Service
- Ticketing
- Bill Estimate
- Field/Logistics
- 3rd Party Integration
- Inventory Management
- Workforce Management
- Service Assurance
- Contracts
- Shopping Cart
- Sales Account Management
- Compensations and Results
- Knowledge Management
- Finance
- Users and Roles
- Collections and Receivable Management

- Latest list and description of each domain can be found [here](#).



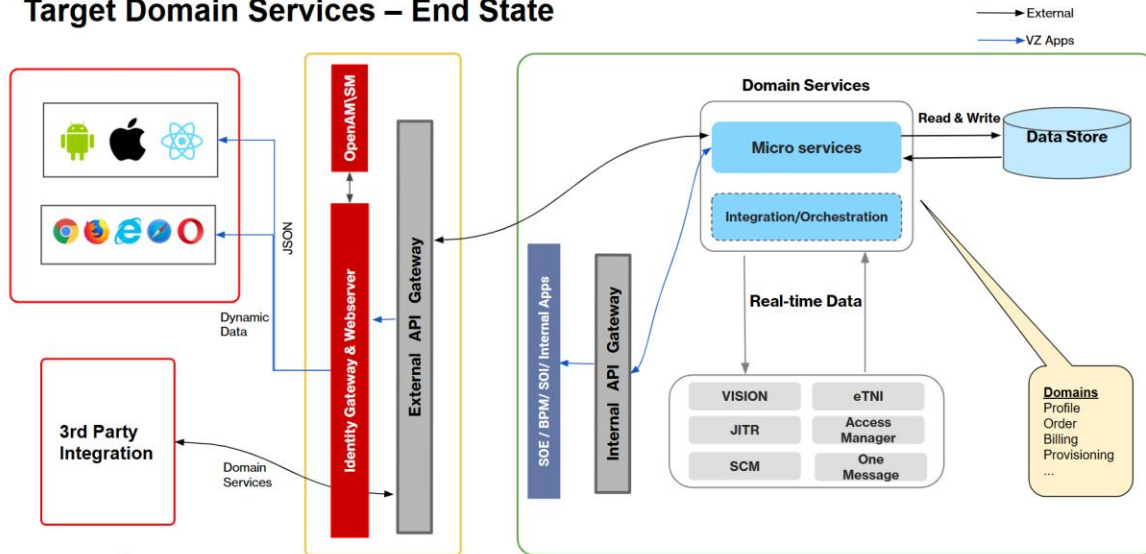
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Domains, Entities & Business Function Services



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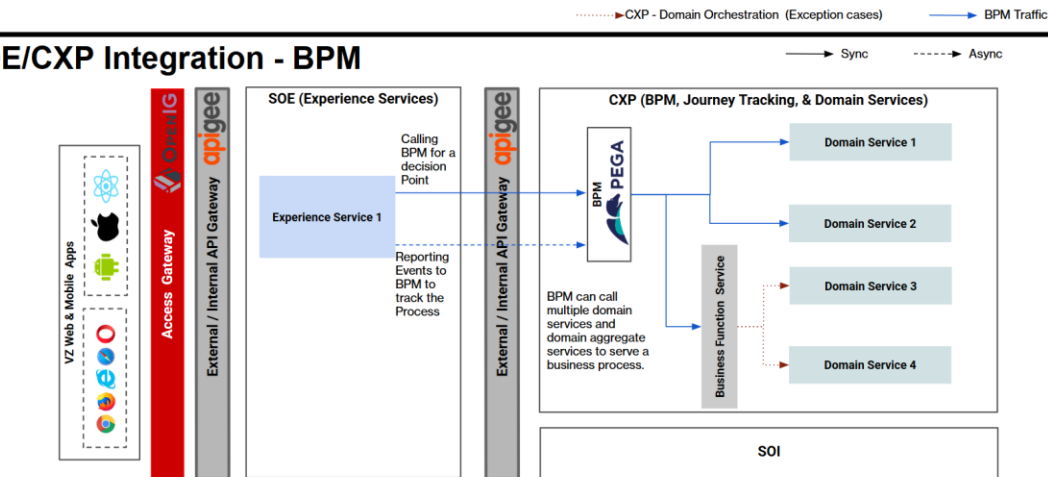
Target Domain Services – End State



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SOE/CXP Integration - BPM

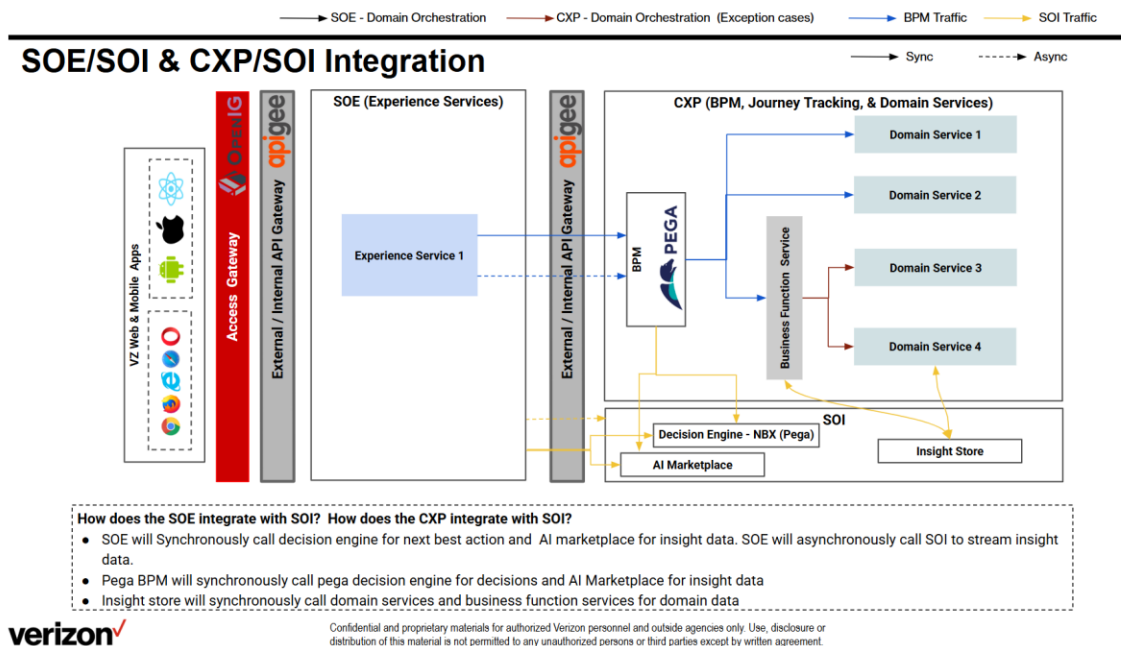


How do Experience Services integrate with BPM for business processes? How does the BPM integrate with domain services and business function services?

- **Async:** Experience Services will asynchronously update the BPM process if there isn't a decision to be made (Tracking the Process)
- **Sync:** Experience Services will synchronously call the BPM process for a decision point (Managing the Process)
- BPM will call domain services and business function services as needed to serve a business process.

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SOE-CXP-SOI Integration - Summary

- **Experience Services → Domain Services & Business Function Services:** Experience Services can orchestrate calls to multiple Domain Services and Business Function Services mainly to retrieve data.
- **Experience Services → BPM:** For Business Processes, Experience Services will integrate with BPM in two patterns:
 - **Async:** Experience Services will asynchronously update the BPM process if there isn't a decision to be made (Tracking the Process)
 - **Sync:** Experience Services will synchronously call the BPM process for a decision point (Managing the Process)
- **Experience Services → SOI:** Experience Services will call SOI to determine the NBX and to determine the pending cases.
- **BPM → Domain Services & Business Function Services:** BPM will call Domain Services & Business Function Services synchronously to retrieve/update domain data.
- **BPM → SOI:** BPM will call SOI synchronously for business insight information.

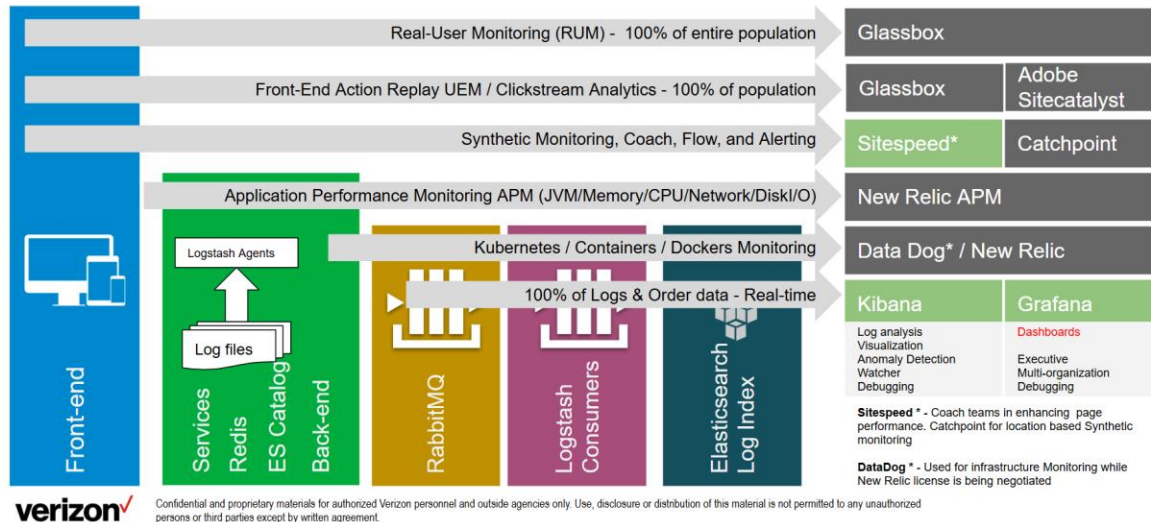


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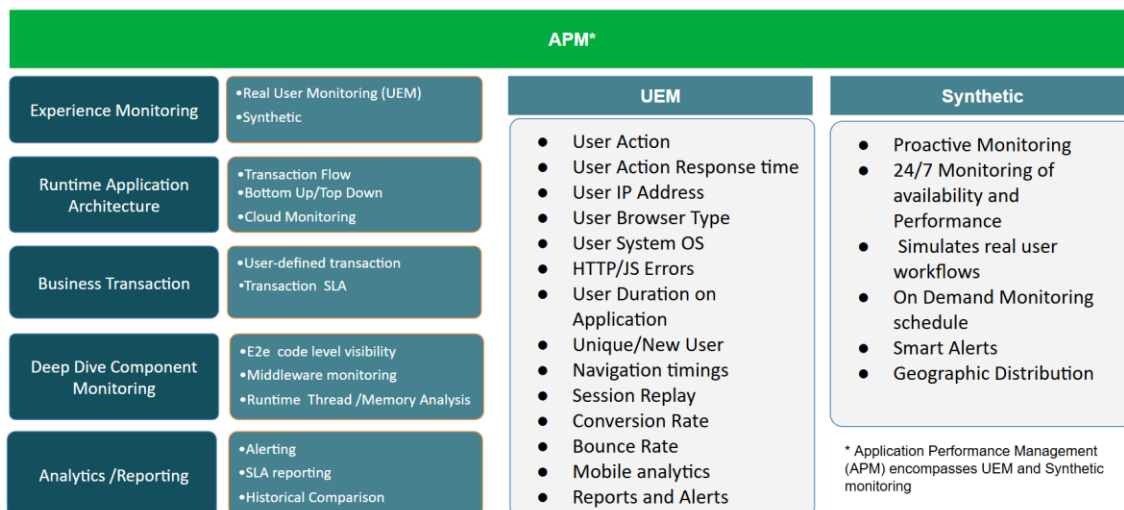
Vendor

Open Source

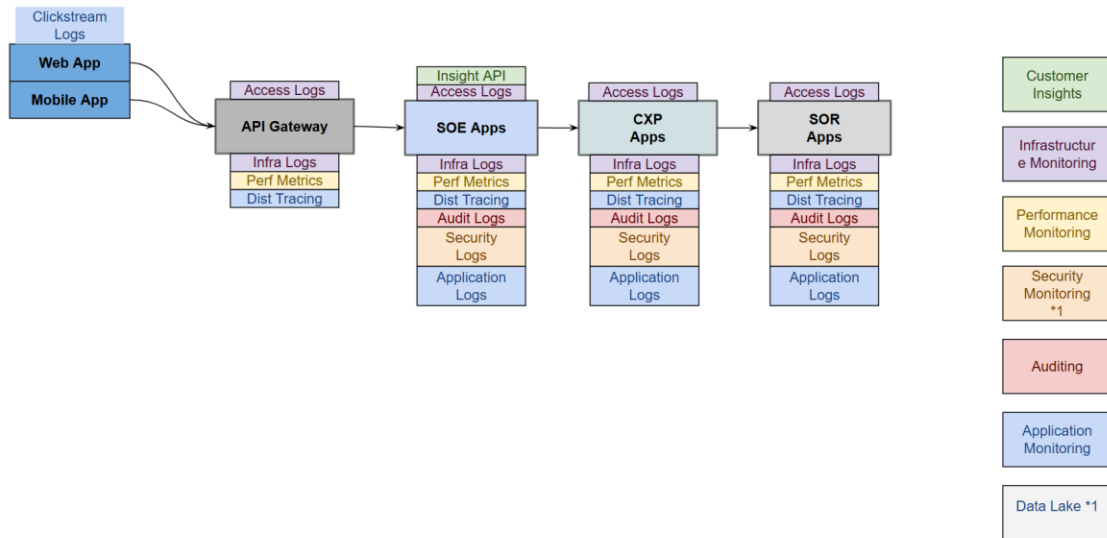
Application Performance Architecture



Application Performance Management



Application Logging & Monitoring

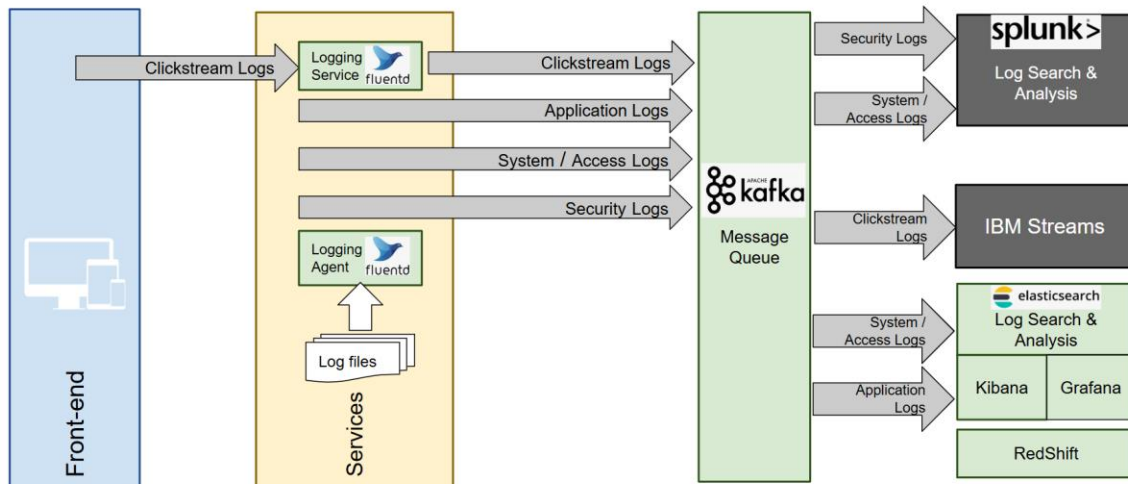


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*1-multiple log sources

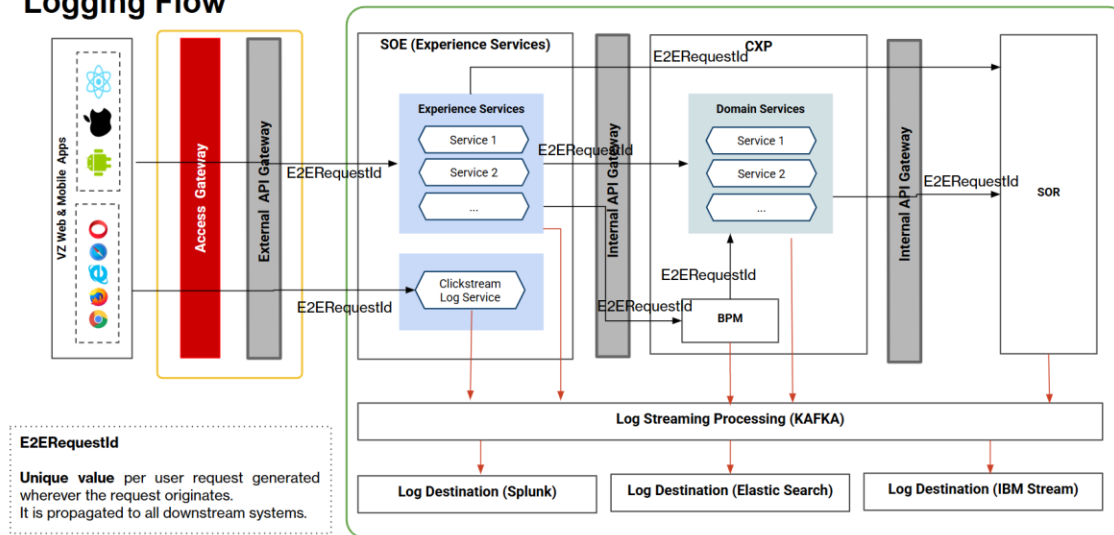
Vendor	Open Source w/ Vendor Support
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Logging Architecture - Components



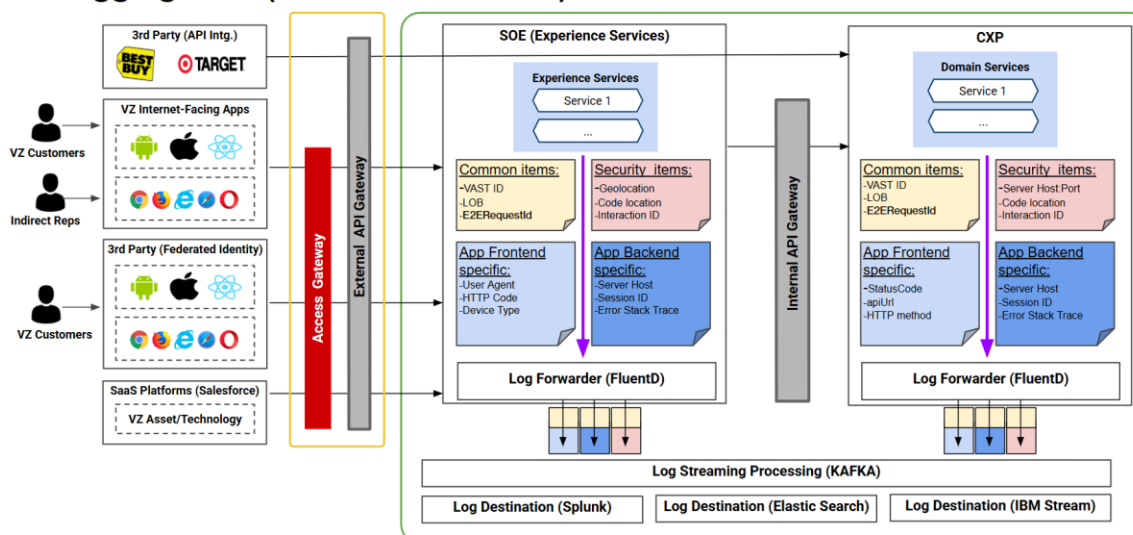
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Logging Flow



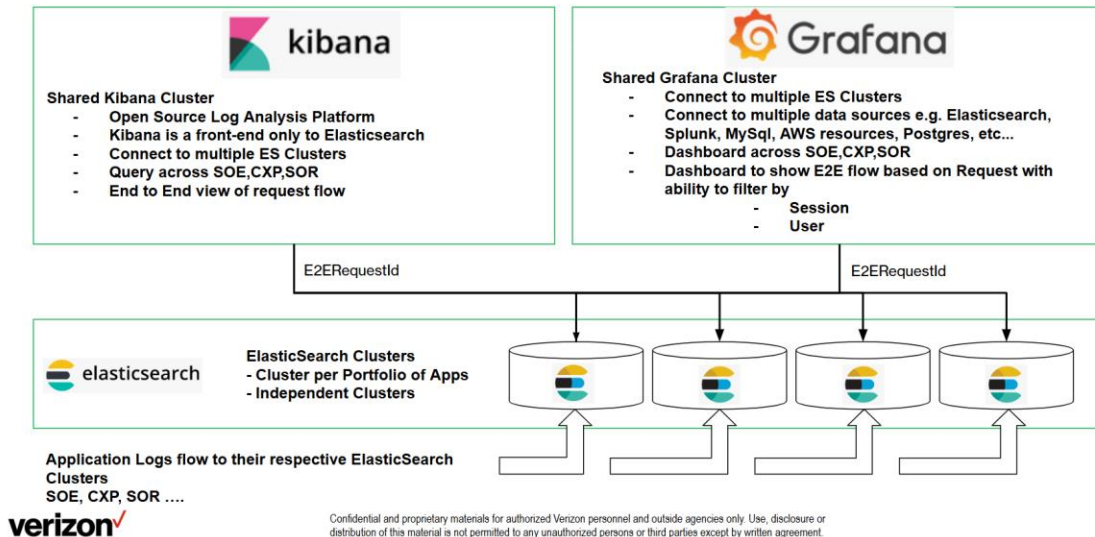
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Logging Flow (Common Structure)

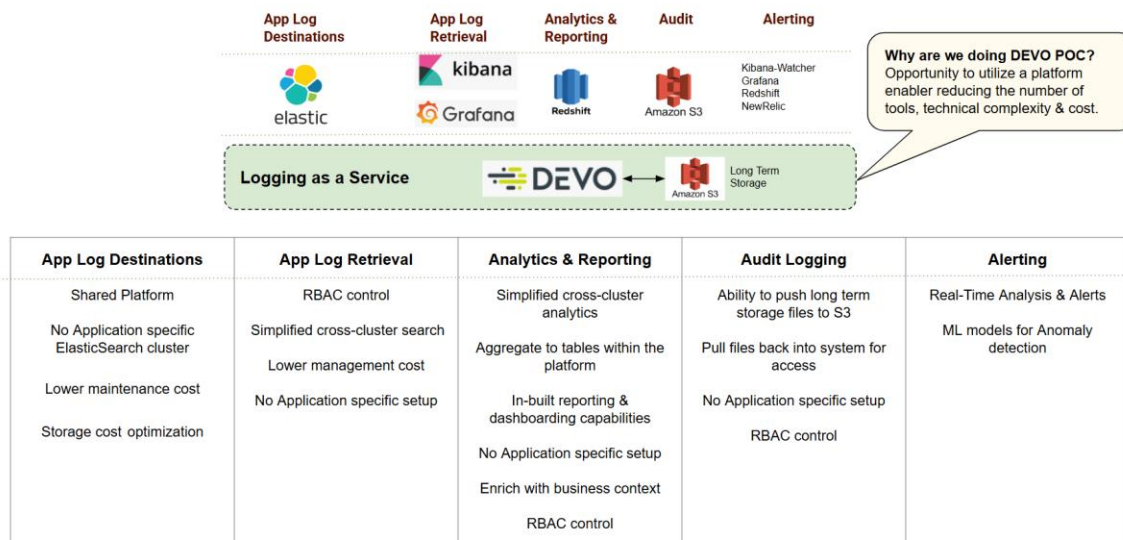


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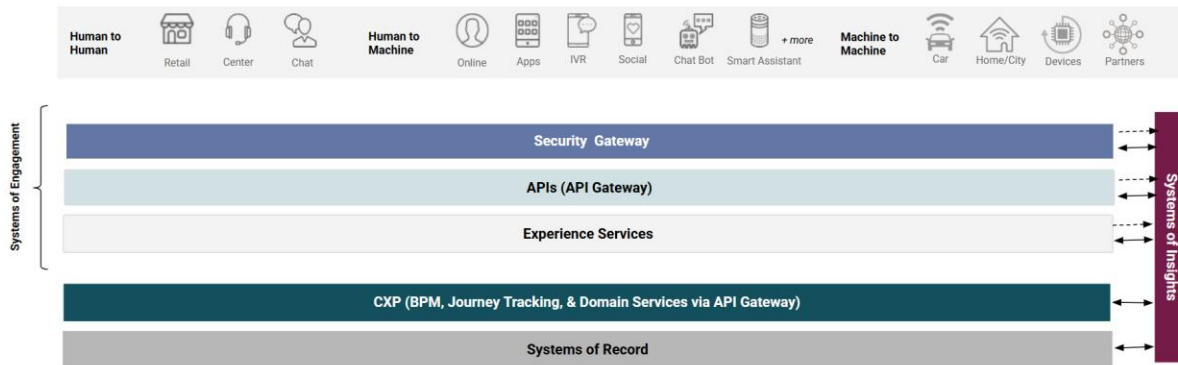
E2E Request View - Application Logs



Devo POC

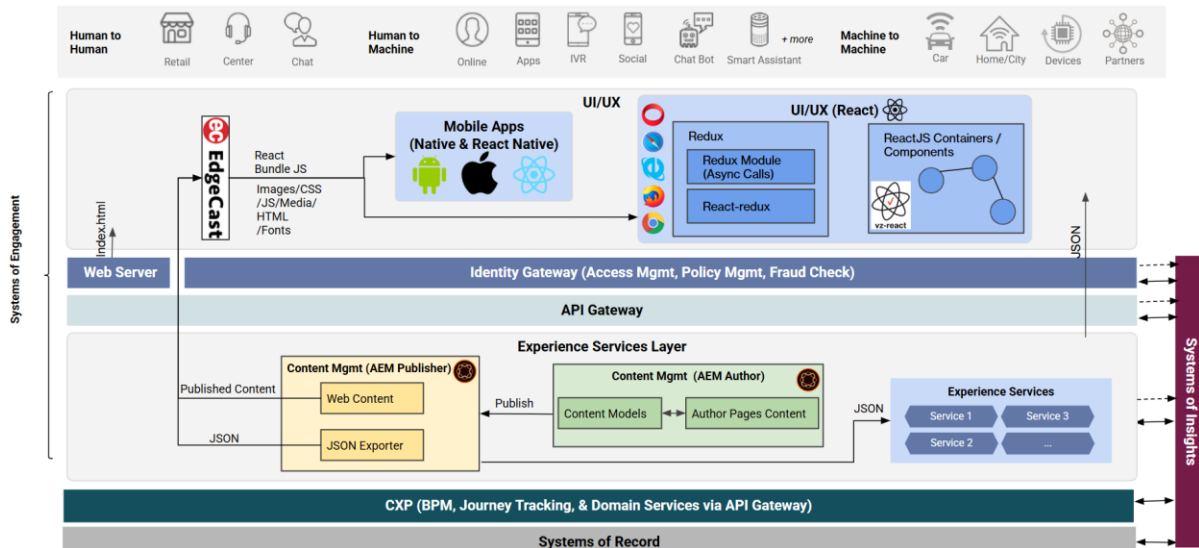


SoE Service Architecture



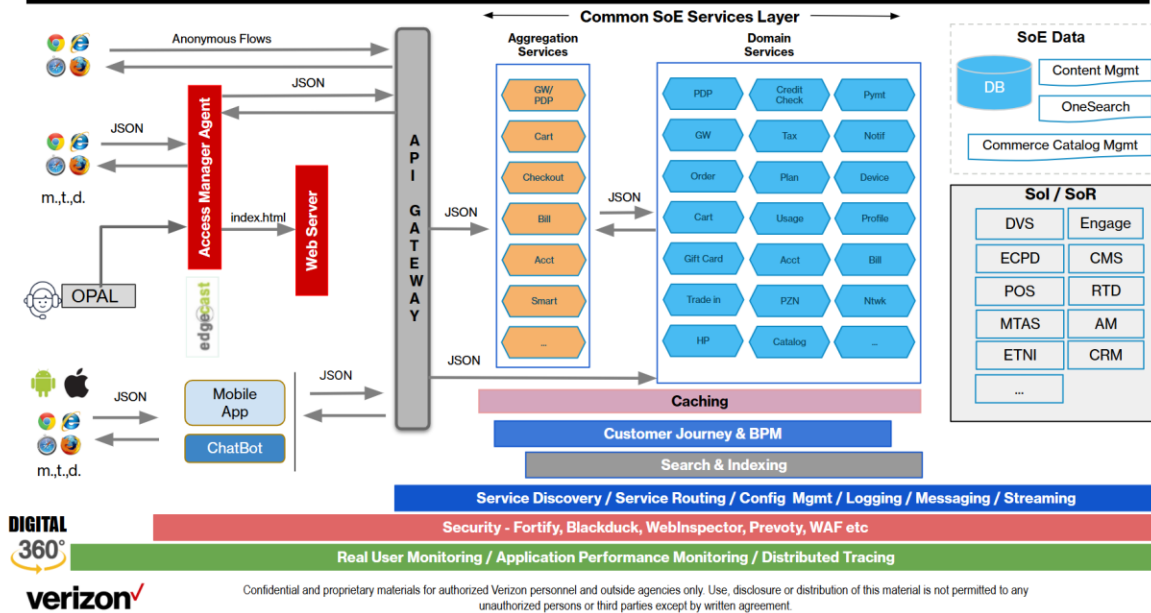
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SoE Front-End Architecture



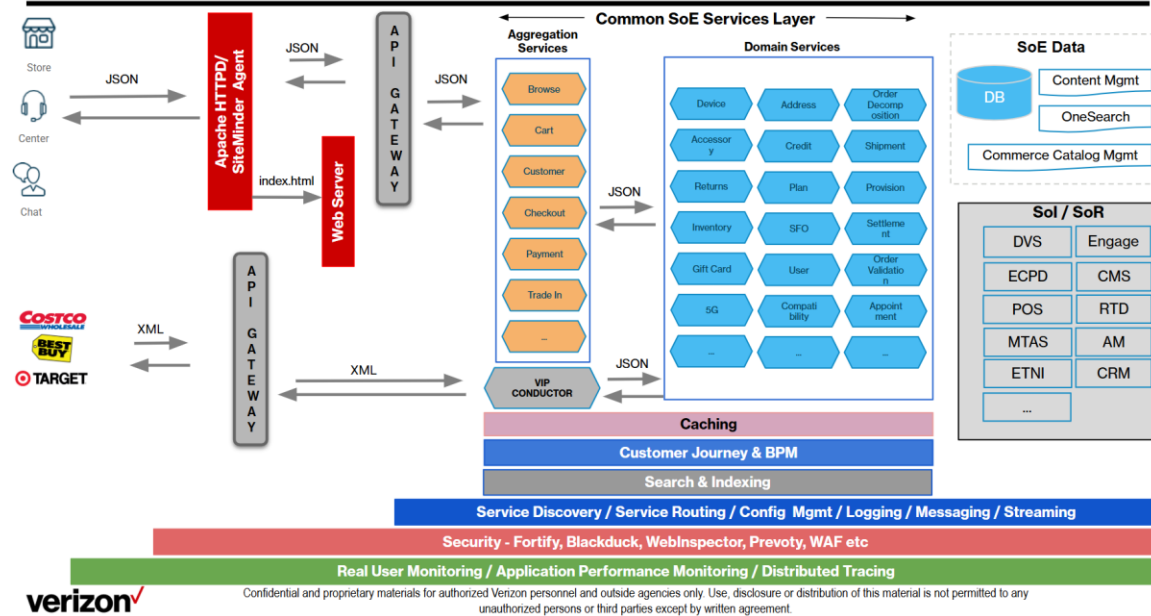
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Target State SoE Digital Architecture - Interactions (Sales & Service)



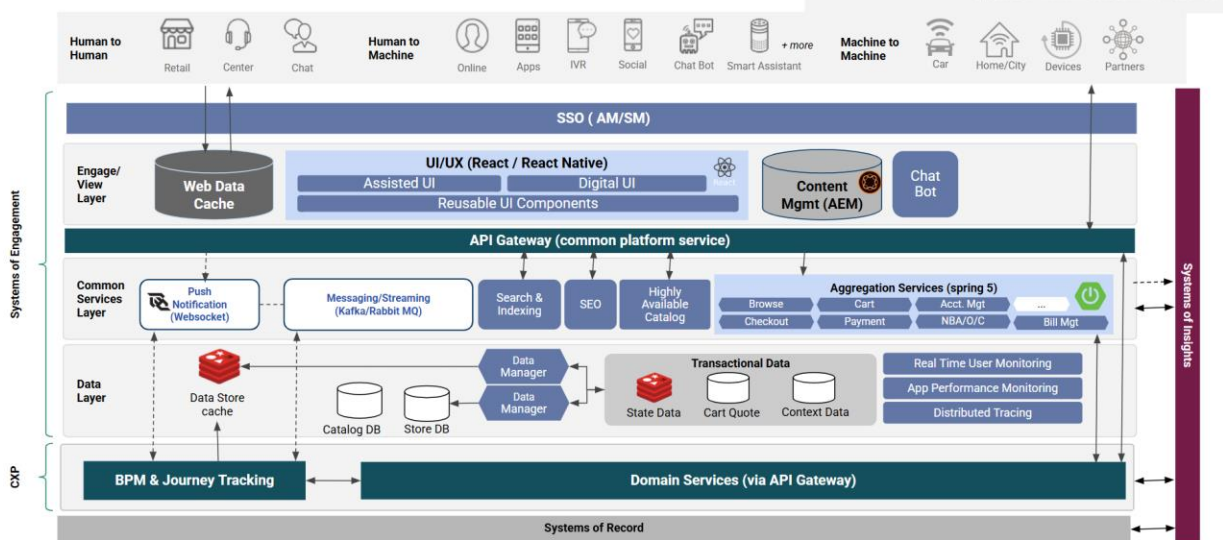
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Target State SoE Assisted Architecture - Interactions (Sales & Service)

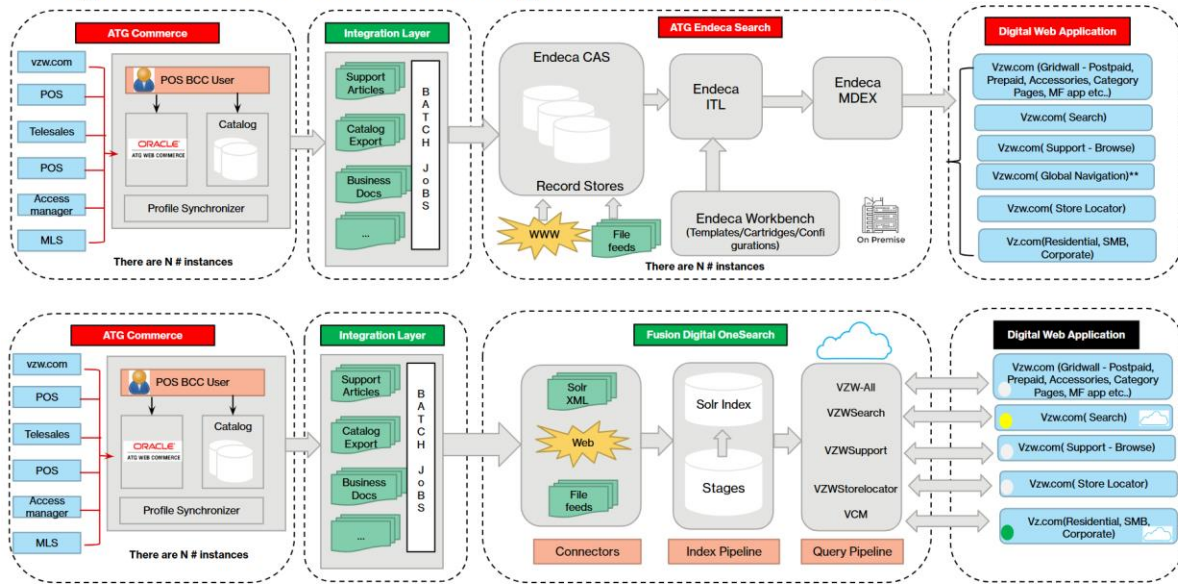


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TO BE UPDATED - Systems of Engagement (Level 1 Architecture)



Digital Endeca to OneSearch Migration Architecture Diagram



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** Migrated to Adobe Experience Manager

Your React Native application may still need an app store update. The following are guidelines of changes to the app that can be supported over the air and those that will require app store updates.

Over-the-air (OTA)

When to use:

- Changes within the same code directory
- Changes to the Javascript code
- Changes to assets like images
- Minor bug fixes and improvements

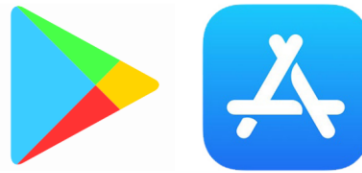
Tools:



App Store Update

When to use:

- Major code changes
- Changes to the Native code
- Adding a 3rd party package
- Upgrades to the React Native package



React Native - Recommendation

React Native is NOT a silver bullet and will not address all your needs. Please evaluate based on your app requirements. React Native is quickly evolving

Development of a NEW Mobile App:

1. Evaluate the Mobile App Requirements
2. Confirm that at least 80% of the App features can be successfully implemented in React Native
3. Develop with React Native and use Native development for components that are not supported by RN yet.

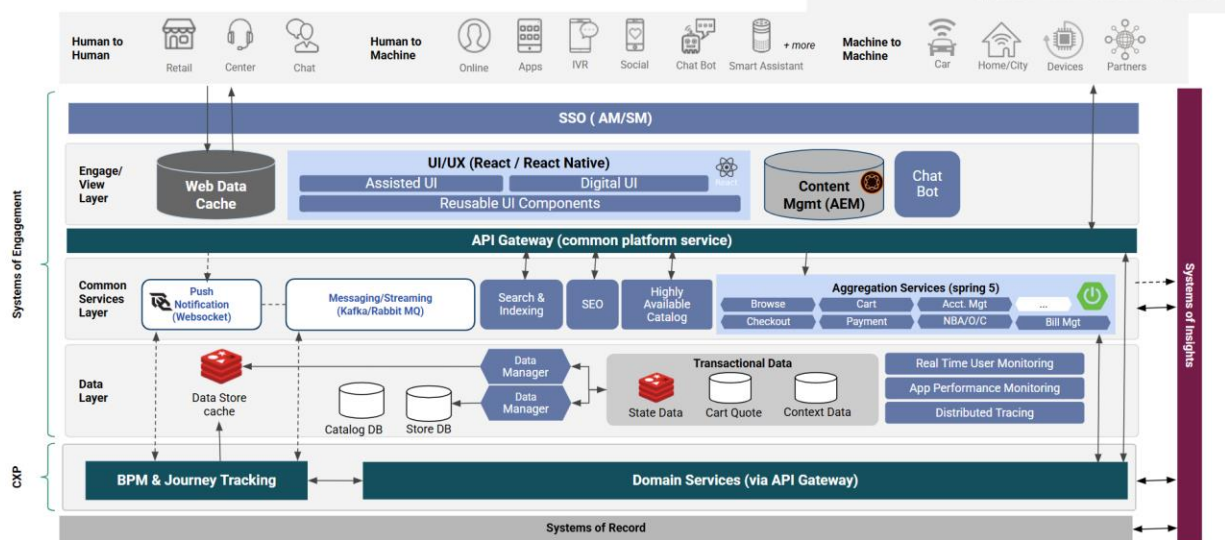
Development of NEW pages or flows in the existing App:

1. Use React Native unless it requires a component that is not supported with RN yet

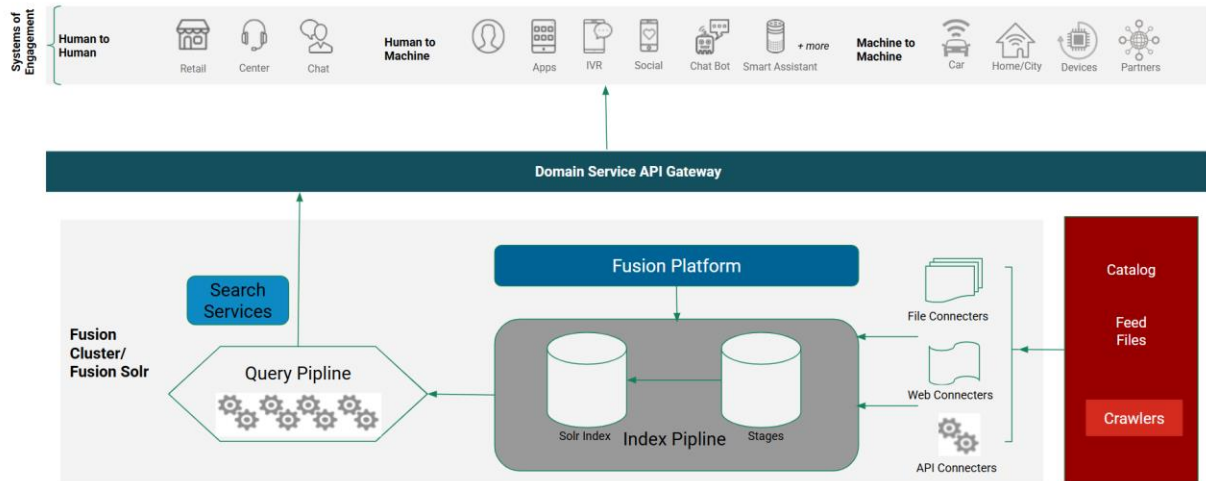
Changes to existing features or pages:

1. Use Native App development until a redesign/refactor is needed.

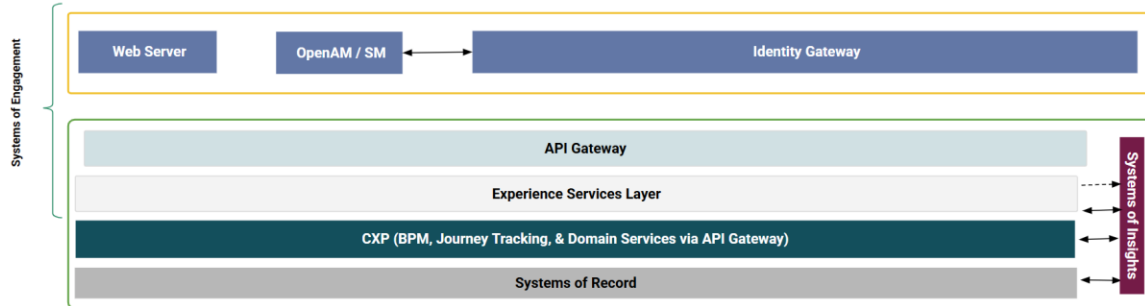
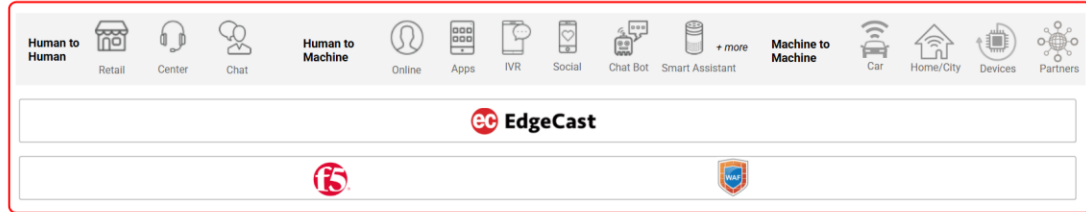
TO BE UPDATED - Systems of Engagement (Level 1 Architecture)



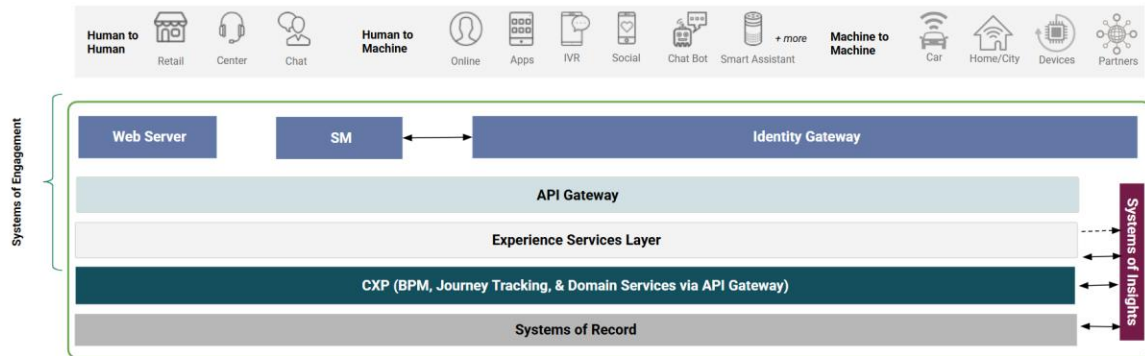
Search (Level 2 Architecture)



SoE Access Mgmt - Digital & Indirect Reps



SoE Access Mgmt - VZ Reps



Access Mgmt - 3rd Party Integrations

